



AutomateX Advisory

Engineering Excellence Amplified

PROPOSAL

InstaTrace V1

Automated Linear Take-off Tool for Cost Estimation

Prepared for	Arcadis Cost Estimating Team
Prepared by	Bill Guo FICE CEng MEng — Founder, AutomateX Advisory
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Contact	bill.guo@automatex.uk automatex.uk

Executive Summary

This proposal sets out AutomateX Advisory's offer to design, build, and deliver InstaTrace V1 — a web-based, AI-native linear take-off tool for the Arcadis Cost Estimating team. V1 automates the manual measurement of linear assets (e.g. access roads, overhead lines) on engineering drawings against the drawing legend, returning a quantified take-off in a fraction of the time required by manual methods.

V1 is scoped tightly around what is on the legend. V2 capabilities (recognition of features not on the legend, land-type intelligence, landmark proximity) are documented as a future development pathway and are explicitly excluded from this proposal.

Headline commercial position

Item	Position
V1 fixed fee	£18,000 + VAT
Payment	30% / 40% / 20% upfront for Phases 1–3; the final 10% (Phase 3 final payment) is invoiced after V1 sign-off. Each upfront phase invoice is issued only after the previous phase gate review is passed.
Delivery	July – September 2026 (Q3)
Beta launch	September 2026, including KPI test on 10 drawings
Maintenance retainer	£800 / month + VAT, quarterly billing in advance (£2,400 per invoice)
Acceptance KPI	≥9 of 10 drawings within ±5% accuracy vs CostX manual take-off
Intellectual property	AutomateX retains code & technology IP; Arcadis has exclusive use of the deployed InstaTrace V1 product.
Hosting	AWS London (eu-west-2); UK-only data residency

The remainder of this document sets out the scope, KPIs, commercial structure, plan, governance, and contractual terms for V1, with a recommended pathway for V2.

1. About AutomateX Advisory

AutomateX Advisory is an AI-native consultancy serving the UK rail, infrastructure, and AEC sector. Our mission is to liberate engineering professionals and business owners from low-value administration so they can focus on high-value judgement and risk management.

Engagement is led by Bill Guo (FICE CEng MEng), founder of AutomateX Advisory, with appropriate supporting resource as required.

2. The Problem & The Opportunity

Linear take-off from engineering drawings is currently a manual, time-intensive task within the Arcadis Cost Estimating team. Estimators perform measurements drawing by drawing, asset by asset, against a drawing legend, typically in CostX. The process carries three structural inefficiencies:

- Time — typical drawings require extended manual tracing and digitisation in CostX that typically take up to 3 weeks per commission.
- Variability — measurements depend on the individual estimator's interpretation and tracing precision.
- Capacity — peak estimate delivery phases concentrate take-off effort in a narrow window, pressing on senior resource availability.

Pilot work conducted by AutomateX has demonstrated that an AI-enabled approach can perform consistent linear take-offs from traced drawings within tight tolerances, returning a structured quantified output in a fraction of the time. InstaTrace V1 productises this capability for use across the Arcadis estimating team.

3. Solution Overview — InstaTrace V1

InstaTrace V1 is delivered as a multi-user web application hosted on AWS UK. There is no desktop software to install. The user workflow is:

1. Upload a PDF engineering drawing (untraced) in the supported V1 format (ArcGIS-derived).
2. Select an asset type from the drawing's legend (e.g. access road, overhead line).
3. The system performs the linear take-off automatically for the selected asset.

4. The system traces the identified asset on the drawing and produces an annotated PDF as output.

5. The user reviews the annotated drawing and downloads the quantified take-off to Excel.

V1 focuses on legend-driven measurement only. The user remains in control: every measurement is annotated visibly on the drawing for review and sign-off before export. The application enforces role-based access for the named user pool, and every measurement is captured in an audit trail (user, timestamp, asset, drawing).

Demo

The capability proposed for V1 has already been demonstrated on a Tier-1 utility drawing during AutomateX's pilot work. The figure below shows automated linear take-off applied to Sheet 20 of the SSEN Beauty to Blackhillock to New Deer to Peterhead 400 kV OHL alignment, with measurements compared against the corresponding manual take-off in CostX.

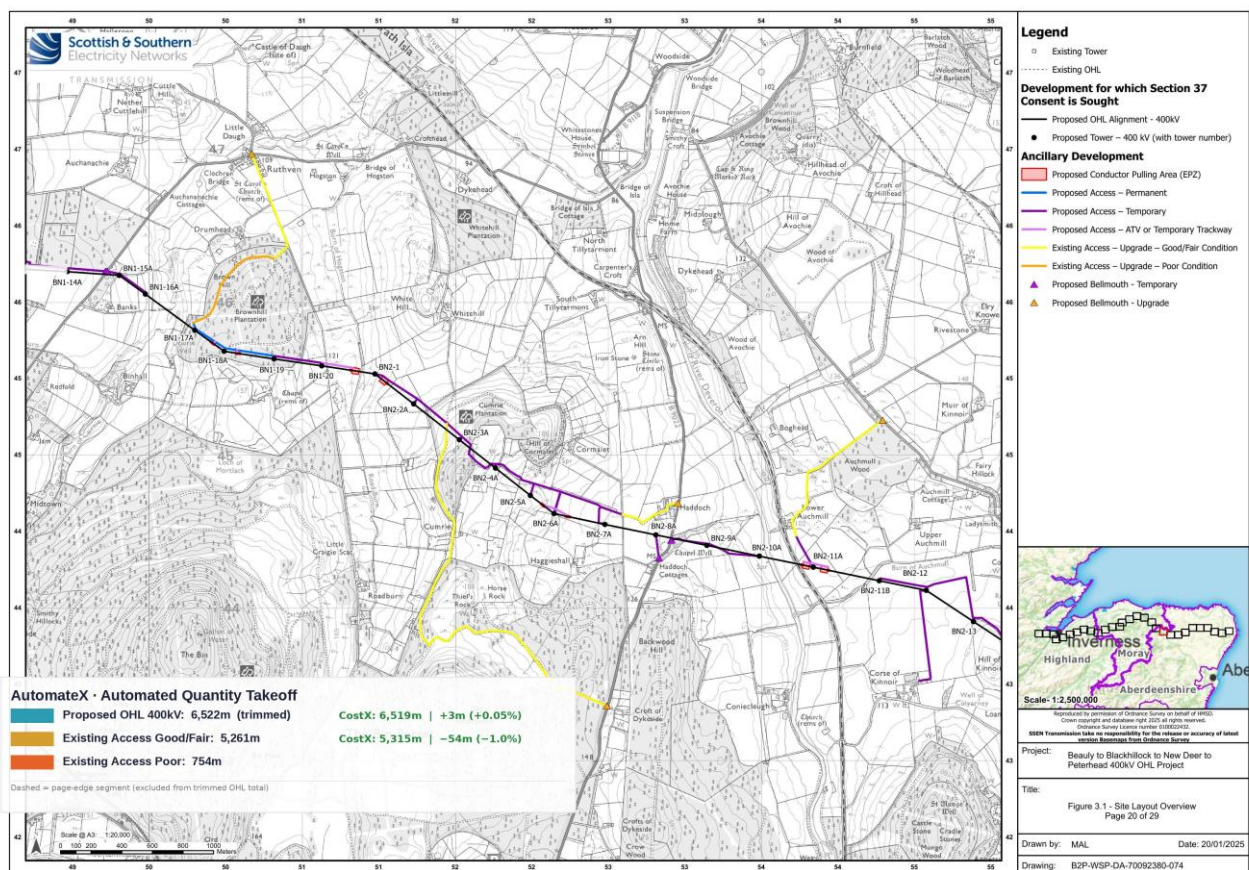


Figure 1 — InstaTrace pilot output on SSEN B2P 400 kV OHL drawing (Sheet 20). Automated take-off achieved +0.05% on Proposed OHL (6,522 m vs CostX 6,519 m) and -1.0% on Existing Access Good/Fair (5,261 m vs CostX 5,315 m), well within the V1 acceptance tolerance of $\pm 5\%$.

4. Scope of V1

4.1 Inclusions

- Web-based application hosted on AWS London (eu-west-2).
- User authentication via email and password, with optional MFA.
- Upload of untraced PDF engineering drawings in the agreed V1 format (ArcGIS-derived) — see Appendix A.
- Legend recognition and asset selection.
- Automated linear take-off for selected legend assets.
- Annotated drawing output (PDF) showing measurements traced for visual review.
- Quantified take-off export to Excel (.xlsx).
- User management for up to 3 named users (additional users available — see Section 6).
- Audit trail of measurements (user, timestamp, asset, drawing).
- Written user manual.
- KPI test on the agreed test pack at beta launch.

4.2 Exclusions — Recommendations for V2

The following capabilities are not included in V1 and are documented here as recommendations for a future V2 phase, subject to a separate proposal:

- Recognition of features “not present” on the drawing legend (e.g. land types, landmarks).
- Land-type overlap analysis against linear assets.
- Distance calculations from linear assets to nominated landmarks.
- Support for drawing formats other than the V1 ArcGIS-derived format.
- Integration with Common Data Environment (CDE) platforms.
- Single Sign-On (e.g. Azure AD).
- Public API for programmatic access.
- Mobile or tablet-optimised interface.

5. Acceptance Criteria & KPIs

V1 sign-off is determined at the end of Phase 3 (September 2026) against the test protocol set out in Appendix A. Successful sign-off closes the project and triggers commencement of the

maintenance retainer. Phase 1 and Phase 2 also have their own lighter gate reviews — these are described in Section 6.1 and govern the issue of the next phase invoice.

The headline KPIs for the V1 sign-off test are:

Metric	Target	Pass condition
Accuracy	±5% per drawing vs CostX manual take-off (ground truth provided by Arcadis)	≥9 of 10 drawings within tolerance
Consistency	Repeatable result on identical drawings	No material variance across re-runs

If the headline accuracy KPI is not met at first KPI test, AutomateX will conduct one further iteration cycle at no additional cost (capped at 5 person-days) based on Arcadis's documented comments. Once Arcadis closes all comments and V1 is signed off, any further changes are managed under Change Control (Section 6.4).

6. Commercial Terms

All amounts are exclusive of VAT. AutomateX Advisory is a UK VAT-registered entity (VAT registration 514 0792 09).

6.1 V1 Build Fee

Total V1 fee: £18,000 + VAT, structured 30% / 40% / 20% / 10% across three sequential phases, with Phase 3 split into a 20% payment at the start of Phase 3 and a final 10% payment after V1 sign-off. The first 90% (£16,200) is paid upfront against gate-review triggers; the final 10% (£1,800) is invoiced only after V1 final acceptance, giving Arcadis assurance that AutomateX retains accountability until sign-off.

Payment logic

- Phases 1–3 are paid upfront. Phase work commences only once the corresponding invoice has been settled.
- The invoice for Phase N is only issued after the Phase N-1 gate review has been passed in writing.
- Phase 1 has no prior gate; its invoice is triggered by PO approval and signed scope.
- The final 10% of the V1 fee (Phase 3 final payment) is invoiced after V1 sign-off — giving Arcadis assurance that AutomateX retains accountability until acceptance.
- Payment terms: 14 days from invoice.

Payment schedule

Phase	Period	%	Amount	Invoice triggered when...
Phase 1	July 2026	30%	£5,400	PO approved & signed scope received
Phase 2	August 2026	40%	£7,200	Phase 1 gate review passed
Phase 3	September 2026	20%	£3,600	Phase 2 gate review passed
Phase 3 — final 10%	Post sign-off	10%	£1,800	V1 final acceptance / sign-off

Phase gate reviews

Gate	At end of	Pass criteria
Phase 1 gate	July 2026	Environment provisioned; authentication working; PDF upload working; legend asset selection demonstrated on a sample drawing
Phase 2 gate	August 2026	Alpha build demonstrating the full V1 user workflow end-to-end on a sample drawing in a controlled review session
Phase 3 deliverable	September 2026	Beta deployed and made available to Arcadis for KPI testing
V1 sign-off	September / early October 2026	KPI test on the 10-drawing pack passed (≥ 9 of 10 within $\pm 5\%$) — see Appendix A. Releases the Phase 3 final 10% payment and commences the maintenance retainer.

V1 sign-off triggers commencement of the maintenance retainer (Section 6.2) and releases the Phase 3 final 10% payment (£1,800 + VAT) for invoice. The first 90% (£16,200) has already been paid upfront in line with the schedule above. If V1 sign-off is not reached at first KPI test, the iteration / remedy process in Section 5 and Appendix A.6 applies; the Phase 3 final 10% payment is withheld until V1 is signed off.

6.2 Maintenance Retainer (post go-live)

£800 / month + VAT, billed quarterly in advance from V1 sign-off (i.e. £2,400 per invoice every 3 months). The retainer covers (1) infrastructure running cost, (2) software maintenance and support, and (3) base usage (3 named users plus 100 take-offs per month). Scope summarised below.

Area	Included in retainer
Hosting & infrastructure	AWS UK hosting, monitoring, and automated backups
Software maintenance	Software updates, dependency upgrades, and security patches
Bug fixing	P1 24 hours initial response; P2 48 hours initial response (UK business hours)
Minor enhancements	Up to approximately 1 person-day per quarter of minor tweaks at AutomateX's discretion
Users	Up to 3 named users

Area	Included in retainer
Usage	Up to 100 take-offs per month, shared across all users (1 take-off = 1 asset measured on 1 drawing — e.g. two asset types on one drawing = 2 take-offs; one asset type on two drawings = 2 take-offs)

Area	Not included — managed under Change Control
New features	Any new features, scope expansion, or UI redesign — including all V2 capabilities (see Section 11)
Drawing formats	Recognition of drawing formats outside the V1 ArcGIS-derived format (V2 — see Section 11)
Integrations	CDE integration, Azure AD SSO, public API (V2 — see Section 11)
Feature recognition	Recognition of features not on the drawing legend (e.g. land types, landmarks); land-type overlap analysis; distance from linear assets to landmarks (V2 — see Section 11)
Platform / device	Mobile or tablet-optimised interface (V2 — see Section 11)

6.3 Additional Users and Usage

Item	Rate
Additional named user	£100 / user / month + VAT
Additional take-offs (50 pack)	£200 / pack + VAT
Additional take-offs (200 pack)	£700 / pack + VAT

Top-up packs do not expire and accumulate in the customer's pool.

Indicative monthly cost scenarios

The figures below are illustrative combinations to help gauge operational cost at different team sizes and usage levels. All figures exclude VAT.

Tier	Users	Takeoffs / mo	Includes	Monthly	Annual
A — Base	3	100	Base retainer only (3 users + 100 takeoffs included)	£800	£9,600

Tier	Users	Takeoffs / mo	Includes	Monthly	Annual
B — Small team	5	200	Base + 2 extra users (£200) + 2 × 50-pack (£400)	£1,400	£16,800
C — Medium team	10	500	Base + 7 extra users (£700) + 2 × 200-pack (£1,400)	£2,900	£34,800
D — Large team	20	1,000	Base + 17 extra users (£1,700) + 4 × 200-pack + 2 × 50-pack (£3,200)	£5,700	£68,400

Where partial pack usage is needed, customers can mix 50-packs (£200) and 200-packs (£700) to tune to expected volume. Unused take-offs do not expire and accumulate in the customer's pool.

6.4 Change Control

Any work outside the V1 scope or retainer envelope is managed under Change Control:

- Day rate: £850 / day + VAT.
- Each change request is scoped, estimated, and agreed in writing before work commences.
- Changes are billed monthly in arrears against actual days delivered.

7. Project Plan & Timeline

The plan below is the calendar view; the underlying gate-review logic that governs payment timing is set out in Section 6.1.

Activity	Window
Proposal review and procurement	May – June 2026
Contract execution and PO	June 2026
Phase 1 — setup & core build	July 2026
Phase 1 gate review	End July 2026
Phase 2 — alpha build	August 2026
Phase 2 gate review	End August 2026
Phase 3 — beta launch & KPI test	September 2026
V1 sign-off	End September / early October 2026
Iteration window (if required)	October 2026
Phase 3 final 10% payment invoiced	On V1 sign-off
Maintenance retainer commences	From V1 sign-off

8. Governance & Roles

Steering

- Phases 1 and 2: fortnightly 30-minute steering call.
- Phase 3 (beta): weekly 30-minute steering call.
- Ad-hoc engagement by exception.

Roles & responsibilities

Side	Role	Responsibility
AutomateX	Founder / Lead (Bill Guo)	Overall delivery, technical lead, primary contact
Arcadis	Project Sponsor	Commercial sign-off, escalation, scope decisions

Side	Role	Responsibility
Arcadis	Product Owner (TBC)	Day-to-day decisions, test pack delivery, beta coordination

9. Assumptions & Dependencies

This proposal is made on the following assumptions:

- Arcadis provides the 10-drawing KPI test pack in the agreed V1 format by the end of Phase 2 (31 August 2026).
- Arcadis provides timely access to a named Product Owner during build and beta.
- Test drawings are representative of the Arcadis estimating team's typical work and conform to the defined V1 format.
- Drawings provided do not contain material defined as Special Category personal data under UK GDPR.
- Arcadis confirms its preferred contracting entity, Data Protection contact, and any vendor security questionnaire requirements during contracting.
- AutomateX delivers InstaTrace V1 as a web-based tool accessible from standard browsers. Arcadis is responsible for any internal IT, network, or security configuration required for its users to access the tool. V1 sign-off is determined by the tool's own functional and accuracy performance — issues outside AutomateX's control (Arcadis IT, network, identity, or other internal infrastructure) do not constitute V1 non-delivery.

10. Terms Summary

Full contractual terms will be set out in a Master Services Agreement and Statement of Work to be issued during contracting. The headline positions are summarised below.

Intellectual property

- AutomateX retains all intellectual property in the underlying technology, including back-end code, models, prompts, and infrastructure-as-code.
- Arcadis is granted exclusive use of the InstaTrace V1 product — the specific deployed instance built under this proposal. No other organisation will be granted access to this instance during the term of the agreement.
- AutomateX may reuse the underlying technology to build separate, derivative products for other customers, deployed on independent infrastructure with no shared users, data, or branding.
- Arcadis retains all rights in the drawings and data it uploads.

Term and termination

- Either party may terminate for convenience on 3 months' written notice.
- Either party may terminate for material breach with 30 days to cure.
- Phase fees paid prior to termination are non-refundable.
- Retainer fees paid in advance are non-refundable; the retainer ends at the end of the notice period.
- AutomateX will provide an export of customer data within 30 days of termination, after which all customer data is securely deleted.

Confidentiality

- Mutual confidentiality obligations to apply, surviving 3 years post termination.
- Drawings and uploaded data treated as Arcadis Confidential Information.

Data protection

- A Data Processing Agreement (Appendix B) forms part of the contract.
- AutomateX acts as Processor; Arcadis as Controller.
- Customer data stored and primarily processed within the United Kingdom (AWS London); AI inference may use sub-processors listed in Appendix B.6 — see DPA for full detail.

11. V2 Pathway

V2 expands InstaTrace beyond the legend to support more sophisticated cost-estimation workflows. V2 proposals are available on request following V1 sign-off; AutomateX will use V1 sign-off feedback to inform V2 scope and pricing. V2 capabilities under consideration include:

- Identification of features not on the drawing legend (land types, landmarks) using a defined symbol library.
- Linear-asset overlap analysis against identified land types.
- Distance calculation from linear assets to nominated landmarks.
- Support for a wider library of drawing formats.
- Optional integrations: CDE, Azure AD SSO, public API.
- Mobile and tablet optimisation.

V2 commercial terms are not included in this proposal and will be agreed separately. The V2 outputs are intended to enrich the linear take-off with intelligence the estimating team can apply commercial judgement against — they are not a replacement for that judgement.

12. Next Steps

To progress, AutomateX proposes the following:

6. Arcadis review of this proposal.

7. Initial follow-up meeting (week commencing 11 May 2026) to walk through the proposal and capture feedback.
8. Arcadis to flag any procurement, security, or legal requirements early to compress contracting.
9. Target contract execution: end of June 2026.
10. Project mobilisation: 1 July 2026.

We look forward to delivering this work and continuing to support Arcadis in deploying AI-native solutions where friction exists in the AEC sector.

For AutomateX Advisory:

Bill Guo FICE CEng MEng

Founder, AutomateX Advisory

bill.guo@automatex.uk | automatex.uk

APPENDIX A

KPI Test Protocol

A.1 Purpose

To define the test conditions under which InstaTrace V1 acceptance is determined.

A.2 Test pack

- 10 drawings, provided by the Arcadis Cost Estimating team.
- All drawings must be in the agreed V1 ArcGIS-derived format.
- Drawings must be representative of the Arcadis estimating team's typical work.
- Each drawing must include a legible legend identifying the linear asset types to be measured.

A.3 Ground truth

- For each drawing in the test pack, Arcadis provides a manual take-off performed in CostX, broken down by asset type.
- Arcadis confirms the manual take-off as the ground truth for the accuracy comparison.

A.4 Test execution

- AutomateX runs InstaTrace V1 against each drawing in the test pack.
- Each take-off produces a quantified result per asset type.
- Results are compared with the CostX ground truth, calculating absolute deviation as a percentage of the ground truth value.

A.5 Pass criteria

- Accuracy: a drawing is considered passed only if every asset type measured on that drawing is within $\pm 5\%$ of the ground truth. If any single asset type on a drawing falls outside $\pm 5\%$, the drawing is counted as failed for the overall pass calculation.
- Overall pass: ≥ 9 of 10 drawings pass.
- Consistency: a re-run of an identical drawing produces a result within $\pm 1\%$ of the prior run.

A.6 Outcome

- Pass: V1 is formally signed off; the Phase 3 final 10% payment (£1,800 + VAT) is invoiced; the project closes; the maintenance retainer commences from the next month.
- Partial or fail: AutomateX undertakes one no-cost iteration cycle based on Arcadis's documented comments. A re-test is then conducted on the same test pack.

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- If the re-test still fails: the parties meet to agree a mutually acceptable remedy in line with the contract.
 - Note: 90% of the V1 fee has already been paid upfront across Phases 1–3 in line with Section 6.1. The KPI test outcome governs V1 sign-off, the release of the Phase 3 final 10% payment (£1,800 + VAT), project closure, and the start of the maintenance retainer.

APPENDIX B

Data Processing Agreement

This Schedule sets out the data protection terms that will form part of the contract between Arcadis (Customer / Controller) and AutomateX Advisory (Supplier / Processor).

B.1 Definitions

Terms in this Schedule have the meanings given to them in UK GDPR and the Data Protection Act 2018. "Customer" means Arcadis (acting as Controller). "Supplier" means AutomateX Advisory (acting as Processor).

B.2 Subject matter and duration

- Subject matter: provision of the InstaTrace V1 software and associated services.
- Duration: for the term of the Agreement, plus 30 days following termination (the data return / deletion window).

B.3 Nature and purpose of processing

- Storage, processing, and analysis of engineering drawings uploaded by Customer for the purpose of automated linear take-off and the production of associated outputs.

B.4 Type of personal data

- Personal data that may incidentally appear within engineering drawings (e.g. landowner names, plot reference identifiers, and addresses appearing on drawings).
- Customer-side user account data: names, business email addresses, and authentication credentials.

B.5 Categories of data subjects

- Landowners and other parties named on uploaded drawings (incidental).
- Customer's authorised users.

B.6 Sub-processors

- AWS (Amazon Web Services EMEA SARL), London region (eu-west-2). Sub-processor for hosting, storage, and backup.
- AI model providers used for inference (currently Anthropic, accessed via API for individual take-off operations). No Customer drawings or data are persisted at the AI provider's infrastructure beyond the duration of the inference call.
- Supplier shall give Customer no less than 30 days' written notice of any proposed change to sub-processors. Customer may object on reasonable data-protection grounds.

B.7 Data location

- All Customer data is stored within the United Kingdom (AWS London).
- AI inference calls used during individual take-off operations may be processed by AI sub-processors listed in B.6 (which may include US-based infrastructure for the duration of the inference call). No Customer data is persisted outside the UK.

B.8 Security measures

- Encryption in transit (TLS 1.2 or higher) and at rest (AES-256).
- Role-based access control with least-privilege principles.
- Regular security patching and dependency management.
- Logged audit trail of administrative access.
- Regular automated backups.
- Cyber Essentials certification is in progress.

B.9 Personal data breach

- Supplier shall notify Customer of any personal data breach affecting Customer data without undue delay, and in any event within 72 hours of becoming aware.
- Supplier shall provide Customer with the information reasonably required to meet Customer's notification obligations under UK GDPR.

B.10 Data subject rights

- Supplier shall, taking into account the nature of the processing, assist Customer by appropriate technical and organisational measures, insofar as possible, in fulfilling Customer's obligations to respond to requests from data subjects.

B.11 Audit

- Customer may request, no more than once per year, evidence of Supplier's compliance with this Schedule.
- Supplier shall respond within 30 days with appropriate evidence (e.g. policy documents, security certificates, audit reports).

B.12 Return and deletion

- On termination, Supplier shall make available to Customer an export of Customer data within 30 days.
- Following the export window, all Customer data held by Supplier shall be securely deleted, save where retention is required by law.

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